RADXA CM5 PROCOMM – COMPLETE ELECTRICAL SCHEMATIC

Page 1 shows real electrical pin/net interconnects. Pages 2–17 contain the complete component-level circuits for PWR–SELECT, CM5–CARRIER, and AUDIO–8X8.

DETAILED ELECTRICAL CIRCUIT PAGES – DOUBLE–CLICK A SHEET TO OPEN

PWR–SELECT
File: ../PWR–SELECT/PowerSelector.kicad_sch

CM5 Interface Contract
File: ../CM5–CARRIER/CM5–Carrier.kicad_sch

CM5 Pin Allocation
File: ../CM5–CARRIER/CM5–Core–Allocated.kicad_sch

Network / PCIe / Wi–Fi
File: ../CM5–CARRIER/Network–PCIe.kicad_sch

WWAN / Dual SIM
File: ../CM5–CARRIER/WWAN–SIM.kicad_sch

Display Harness
File: ../CM5–CARRIER/Display–Harness.kicad_sch

Headset / Audio Control
File: ../CM5–CARRIER/Audio–Control.kicad_sch

Power Regulators
File: ../CM5–CARRIER/Power–Regulators–A1.kicad_sch

Thermal / Fans / IO
File: ../CM5–CARRIER/Thermal–IO.kicad_sch

AUDIO–8X8 Interface
File: ../AUDIO–8X8/Audio–8x8.kicad_sch

TDM / Clock / Control
File: ../AUDIO–8X8/Audio–TDM–Clock.kicad_sch

AK5558VN ADC
File: ../AUDIO–8X8/AK5558–ADC.kicad_sch

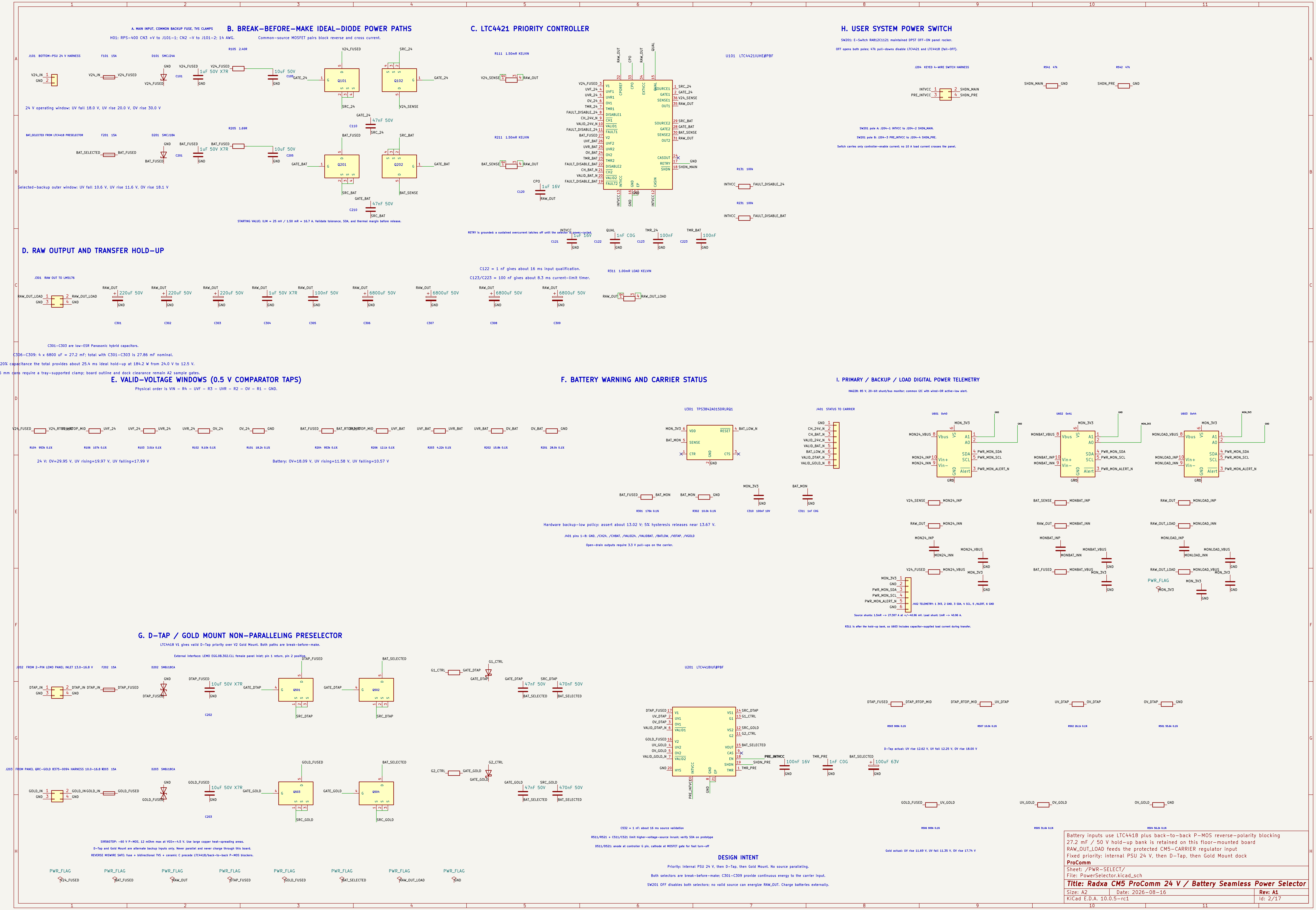
AK4458VN DAC
File: ../AUDIO–8X8/AK4458–DAC.kicad_sch

Balanced Inputs 1–8
File: ../AUDIO–8X8/Audio–Inputs.kicad_sch

Balanced Outputs 1–8
File: ../AUDIO–8X8/Audio–Outputs.kicad_sch

Audio Power
File: ../AUDIO–8X8/Audio–Power.kicad_sch

PCB ROUTING HELD FOR PHYSICAL RELEASE GATES
~~System connector representations are excluded from BOM and board update~~
Three physical PCB netlists remain separate
Top sheet is the electrical system interconnect; pages 2–17 contain detailed circuits
ProComm
Sheet: /
File: Radxa–CM5–ProComm–System.kicad_sch
Title: Radxa CM5 ProComm – Complete Electrical Schematic
Size: A0 | Date: 2026–08–17 | **Rev: A2**
KiCad E.D.A. 10.0.5–rc1 | Id: 1/17



A. MAIN INPUT, COMMON BACKUP FUSE, TVS CLAMPS
B. BREAK-BEFORE-MAKE IDEAL-DIODE POWER PATHS
H01: RPS-400 CN3 +V to J101-1; CN2 -V to J101-2; 14 AWG.
Common-source MOSFET pairs block reverse and cross current.
24 V operating window: UV fall 18.0 V, UV rise 20.0 V, OV rise 30.0 V
Selected-backup outer window: UV fall 10.6 V, UV rise 11.6 V, OV rise 18.1 V
STARTING VALUE: ILIM = 25 mV / 1.50 mR = 16.7 A. Validate tolerance, SOA, and thermal margin before release.
C. LTC4421 PRIORITY CONTROLLER
U101 LTC4421IUHE#PBF
RETRY is grounded: a sustained overcurrent latches off until the selector is power-cycled.
C122 = 1 nF gives about 16 ms input qualification.
C123/C223 = 100 nF gives about 8.3 ms current-limit timer.
R311 1.00mR LOAD KELVIN
H. USER SYSTEM POWER SWITCH
SW201: E-Switch RA812C1121 maintained DPST OFF-ON panel rocker.
OFF opens both poles; 47k pull-downs disable LTC4421 and LTC4418 (fail-OFF).
J204 KEYED 4-WIRE SWITCH HARNESS
SW201 pole A: J204-1 INTVCC to J204-2 SHDN_MAIN.
SW201 pole B: J204-3 PRE_INTVCC to J204-4 SHDN_PRE.
Switch carries only controller-enable current; no 10 A load current crosses the panel.
D. RAW OUTPUT AND TRANSFER HOLD-UP
J301 RAW OUT TO LMS176
C301-C303 are low-ESR Panasonic hybrid capacitors.
C306-C309: 4 x 6800 uF = 27.2 mF; total with C301-C303 is 27.86 mF nominal.
At -20% capacitance the total provides about 25.4 ms ideal hold-up at 184.2 W from 24.0 V to 12.5 V.
x 35 mm cans require a tray-supported clamp; board outline and dock clearance remain A2 sample gates.
E. VALID-VOLTAGE WINDOWS (0.5 V COMPARATOR TAPS)
Physical order is VIN - R4 - UVF - R3 - UVR - R2 - OV - R1 - GND.
24 V: OV=29.95 V, UV rising=19.97 V, UV falling=17.99 V
Battery: OV=18.09 V, UV rising=11.58 V, UV falling=10.57 V
F. BATTERY WARNING AND CARRIER STATUS
U301 TPS3842A015DRLRQ1
J401 STATUS TO CARRIER
Hardware backup-low policy: assert about 13.02 V; 5% hysteresis releases near 13.67 V.
J401 pins 1-8: GND, /CH24, /CHBAT, /VALID24, /VALIDBAT, /BATLOW, /VDTAP, /VGOLD
Open-drain outputs require 3.3 V pull-ups on the carrier.
I. PRIMARY / BACKUP / LOAD DIGITAL POWER TELEMETRY
INA228: 85 V, 20-bit shunt/bus monitor; common I2C with wired-OR active-low alert.
J402 TELEMETRY: 1 3V3, 2 GND, 3 SDA, 4 SCL, 5 /ALERT, 6 GND
R311 is after the hold-up bank, so U603 includes capacitor-supplied load current during transfer.
G. D-TAP / GOLD MOUNT NON-PARALLELING PRESELECTOR
LTC4418 V1 gives valid D-Tap priority over V2 Gold Mount. Both paths are break-before-make.
External interface: LEMO EGG.0B.302.CLL female panel inlet; pin 1 return, pin 2 positive.
J202 FROM 2-PIN LEMO PANEL INLET 13.0-16.8 V
J203 FROM PANEL QRC-GOLD 8375-0094 HARNESS 10.0-16.8 V
U201 LTC4418IUF#PBF
D-Tap actual: UV rise 12.62 V, UV fall 12.25 V, OV rise 18.00 V
Gold actual: UV rise 11.69 V, UV fall 11.35 V, OV rise 17.74 V
SIR5607DP: -60 V P-MOS, 12 mOhm max at VGS=-4.5 V. Use large copper heat-spreading areas.
D-Tap and Gold Mount are alternate backup inputs only. Never parallel and never charge through this board.
REVERSE MISWIRE SAFE: fuse + bidirectional TVS + ceramic C precede LTC4418/back-to-back P-MOS blockers.
C532 = 1 nF: about 16 ms source validation
R511/R521 + C511/C521 limit higher-voltage-source inrush; verify SOA on prototype
D511/D521: anode at controller G pin, cathode at MOSFET gate for fast turn-off
DESIGN INTENT
Priority: internal PSU 24 V, then D-Tap, then Gold Mount. No source paralleling.
Both selectors are break-before-make; C301-C309 provide continuous energy to the carrier input.
SW201 OFF disables both selectors; no valid source can energize RAW_OUT. Charge batteries externally.
Battery inputs use LTC4418 plus back-to-back P-MOS reverse-polarity blocking
27.2 mF / 50 V hold-up bank is retained on this floor-mounted board
RAW_OUT_LOAD feeds the protected CM5-CARRIER regulator input
Fixed priority: internal PSU 24 V, then D-Tap, then Gold Mount dock
ProComm
Sheet: /PWR-SELECT/
File: PowerSelector.kicad_sch
Title: Radxa CM5 ProComm 24 V / Battery Seamless Power Selector
Size: A2
Date: 2026-08-16
Rev: A1
KiCad E.D.A. 10.0.5-rc1
Id: 2/17

A. PWR–SELECT INTERFACE

RAW_POWER_FROM_PWR_SELECT — J101

RAW_OUT_LOAD	1	2	RAW_OUT_LOAD
GND	3	4	GND

Mates with PWR–SELECT J301; two contacts per polarity; validate 15 A derating.

PWR_STATUS_FROM_SELECTOR — J102

GND	1
CH_24V_N	2
CH_BAT_N	3
VALID_24V_N	4
VALID_BAT_N	5
BAT_LOW_N	6
VALID_DTAP_N	7
VALID_GOLD_N	8

Mates with PWR–SELECT J401; add 3.3 V pull–ups on the carrier.

PWR_TELEMETRY_FROM_SELECTOR — J103

LOGIC_3V3	1
GND	2
CTRL_I2C_SDA	3
CTRL_I2C_SCL	4
PWR_MON_ALERT_N	5
GND	6

Mates with PWR–SELECT J402; carrier supplies 3.3 V and owns I2C pull–ups.

B. BUFFERED DIFFERENTIAL TDM + AUDIO CONTROL

AUDIO_TDM_CONTROL — J201

AUD_MCLK_P	1	2	AUD_MCLK_N
GND	3	4	GND
AUD_BCLK_P	5	6	AUD_BCLK_N
AUD_FSYNC_P	7	8	AUD_FSYNC_N
GND	9	10	GND
AUD_DAC_SDIN_P	11	12	AUD_DAC_SDIN_N
AUD_ADC_SDOUT_P	13	14	AUD_ADC_SDOUT_N
GND	15	16	GND
AUD_I2C_SCL	17	18	AUD_I2C_SDA
AUD_ADC_RST_N	19	20	AUD_DAC_RST_N
AUD_DAC_MUTE_N	21	22	AUD_IRQ_N
AUDIO_PRESENT_N	23	24	AUDIO_ENABLE
LOGIC_3V3	25	26	GND
TDM_SPARE_1	27	28	TDM_SPARE_2
GND	29	30	GND

AUDIO_12V_POWER — J202

AUDIO_12V	1	2	AUDIO_12V
GND	3	4	GND

Separate filtered audio–power harness; no fan/radio loads.

Mates with AUDIO–8X8 J101 using Molex 87832–6423 headers; harness assembly remains vibration–qualified.

Carrier drives MCLK/BCLK/FSYNC/DAC data; AUDIO–8X8 drives ADC data back.

C. FOUR INDEPENDENT PWM / TACH FANS

CPU_FAN — J401

GND	1	2	CPU_FAN_12V
CPU_FAN_TACH	3	4	CPU_FAN_PWM

CPU_FAN

MODEM_FAN — J402

GND	1	2	MODEM_FAN_12V
MODEM_FAN_TACH	3	4	MODEM_FAN_PWM

MODEM_FAN

THA0412AD–TZW3_INTAKE — J403

GND	1	2	INTAKE_FAN_12V
INTAKE_FAN_TACH	3	4	INTAKE_FAN_PWM

THA0412AD–TZW3_INTAKE

THA0412AD–TZW3_EXHAUST — J404

GND	1	2	EXHAUST_FAN_12V
EXHAUST_FAN_TACH	3	4	EXHAUST_FAN_PWM

THA0412AD–TZW3_EXHAUST

J403 is physically keyed/labeled INTAKE; J404 is keyed/labeled EXHAUST.

D. DETAILED CAPTURE SUITE – REV A1

01 CM5–Core–Allocated: exact 300–contact CM5 symbol, 76 locked functions

02 CM5–Core–Allocated: VCC_SYSIN, reset, recovery, power–on and debug UART

03 Network–PCIe: PI7C9X2G608GP, 3x LAN7430 and native WAN1 front end

04 Network–PCIe: 4x Wurth MagJack and Molex 0679101002 Mini PCIe 4T4R Wi–Fi

05 WWAN–SIM: TE 2199230–3 B–key, USB3/USB2, controls and dual–SIM mux

06 Display–Harness: HDMI, USB touch and dedicated 12 V / 2.5 A branch

07 Audio–Control: I2S0 TDM, I2S1 ES8316 headset and CTIA amplifier path

08 Thermal–IO: I2C translation, GPIO expansion, sensors and four fan channels

09 Power–Regulators–A1: protected raw hold–up, all main rails, sequencing and test points

10 Audio–8x8: all XLR shields bonded to chassis with one controlled AGND bond

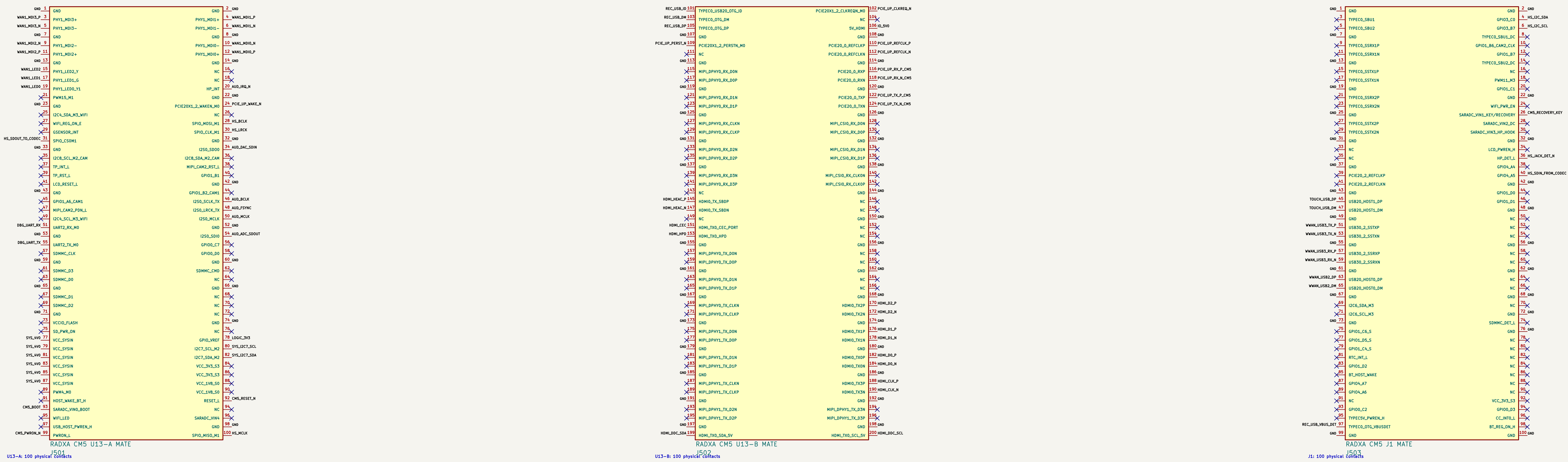
11 Review: every detailed sheet has zero ERC errors; warnings are classified

DC/DC compensation and magnetics remain a separate power–design calculation milestone.

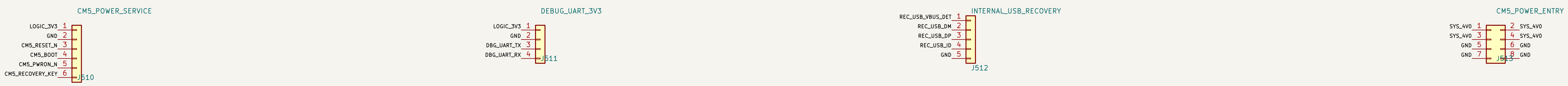
High–speed routing starts only after stackup and connector Z datums are released.

The two enclosure fans require an underside baffle to prevent direct intake/exhaust short flow.

1. EXACT RADXA CM5 MATING CONNECTORS



2. INTERNAL POWER / STARTUP / RECOVERY SERVICE



3.3 V UART only; keyed factory/debug cable.

Internal keyed recovery harness; service fixture owns the protected USB-C receptacle and CC network.

Kelvin-sense SYS_4V0 at J501; 4.0 V follows the Radxa CM5 carrier design note.

Internal keyed service header; never exposed on the top panel.

3. OWNERSHIP AUDIT

76 / 76 allocated contacts connected; 0 duplicate physical-pin claims.

WAN1 uses native MDI. PCIe Gen2 x1 feeds the packet switch.

WWAN owns USB30_2 + USB20_HOST0; touch owns USB20_HOST1.

HDMI_TX0, I2S0 TDM, I2S1 headset, I2C7, I2C3 and UART2 are reserved exactly once.

Pins 145/147 remain HDMI HEAC reserved. U13-B pin 106 receives the dedicated IO_5V0 rail.

Unused alternate-function contacts carry explicit no-connect markers on this A1 baseline.

DETAILED CAPTURE BASELINE - verify CM5 voltage limits before release
All connector grounds join the carrier ground plane; allocated GPIO domain is 3.3 V
Exactly 76 product allocations are connected; unused functions are intentionally no-connect
Three 100-contact Hirose DF40C mates; all physical pin numbers follow CM5 V2.21

ProComm

Sheet: /CM5 Pin Allocation/
File: CM5-Core-Allocated.kicad_sch

Title: Radxa CM5 ProComm Carrier - Exact CM5 Connector Allocation

Size: A2	Date: 2026-08-13	Rev: A1
KiCad E.D.A. 10.0.5-rc1		Id: 4/17

Port 5 is disabled and unrouted; generic cable headers are prohibited on PCIe Gen2 lanes
Exact package balls/pins from Diodes DS40210 Rev 8 and Microchip DS00002631G
Native WAN1 plus three LAN7430 endpoints use exact Wurth 74991114412 MagJacks
PI7C9X2G608GP in 606 mode: x1 upstream plus five x1 downstream ports

ProComm

Sheet: /Network / PCIe / Wi-Fi/
File: Network-PCIe.kicad_sch

Title: Radxa CM5 ProComm Carrier — PCIe / Ethernet / Wi-Fi

Size: A1	Date: 2026-08-14	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 5/17

RF coax leaves the module directly for four right-side bulkhead antennas
FSA2567 defaults to physical SIM 1; open-drain low selects physical SIM 2
Dedicated 3.8 V / 5 A minimum rail; verify selected modem peak-current waveform
M.2 B-key modem socket supports USB 3.0 and USB 2.0 fallback

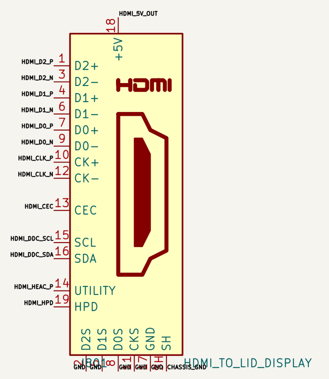
ProComm

Sheet: /WWAN / Dual SIM/
File: WWAN-SIM.kicad_sch

Title: Radxa CM5 ProComm Carrier – WWAN and Dual SIM

Size: A1	Date: 2026-08-14	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 6/17

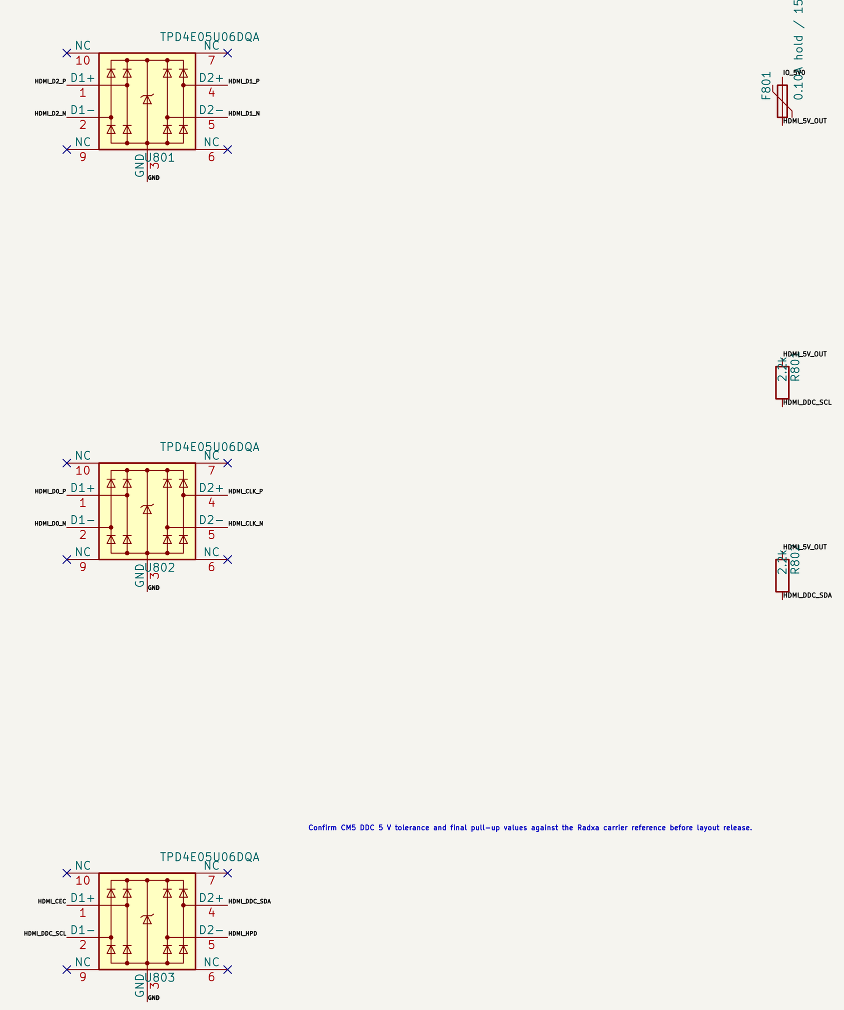
1. HDMI TYPE-A SOURCE CONNECTOR



HDMI_HEAC_N remains unused in Rev A; the selected monitor uses video/DDC/CEC/HPD only.

Shell bonds to chassis at the connector. TMDS shield returns use the local digital ground plane.

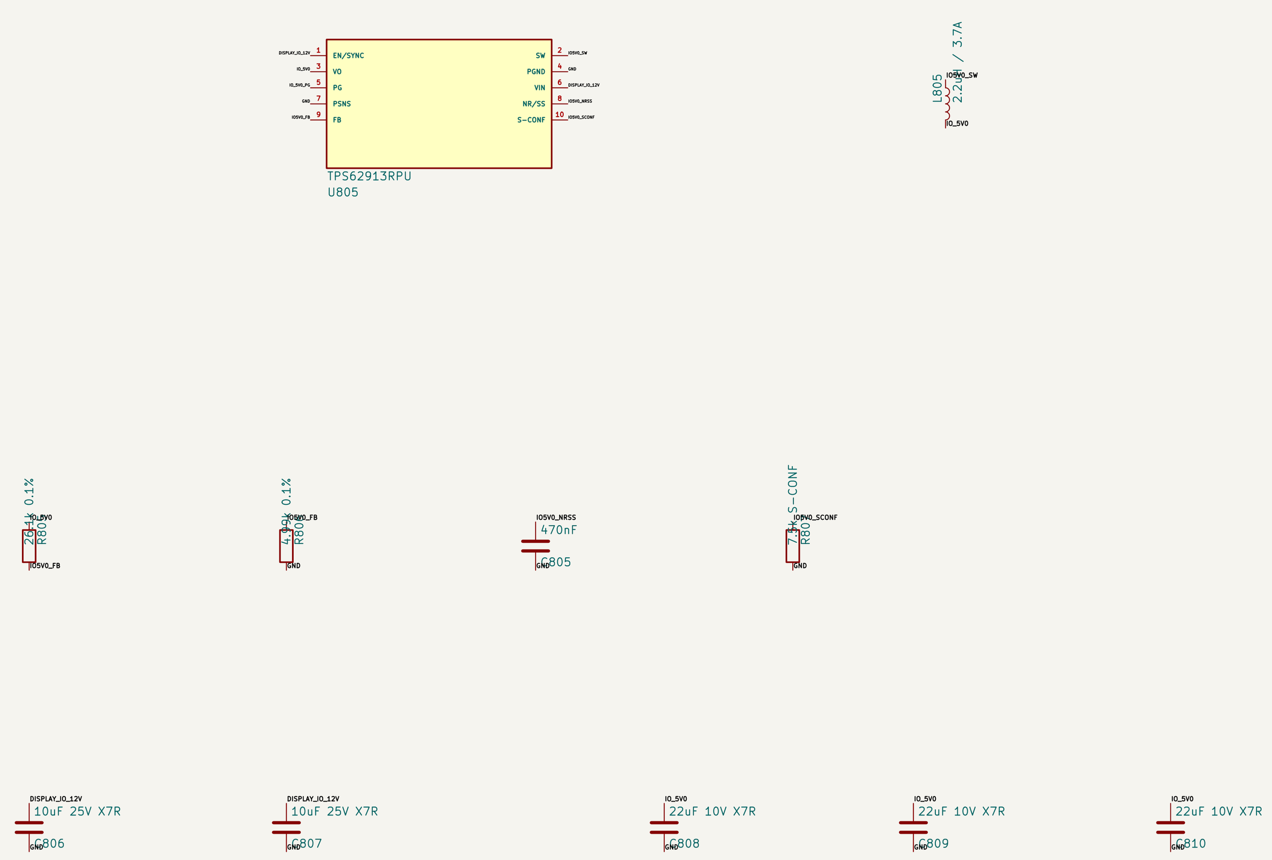
2. HDMI ESD / LOW-CAPACITANCE SHUNTS



Confirm CM5 DDC 5 V tolerance and final pull-up values against the Radxa carrier reference before layout release.

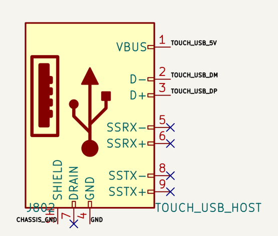
Place all HDMI ESD arrays immediately behind J801 with no stubs on TMDS pairs.

3. DEDICATED IO_5V0 / 2 A BUCK



IO_5V0 feeds CM5 U13-B pin 106, HDMI source 5 V and USB-touch VBUS; no monitor-power eFuse.

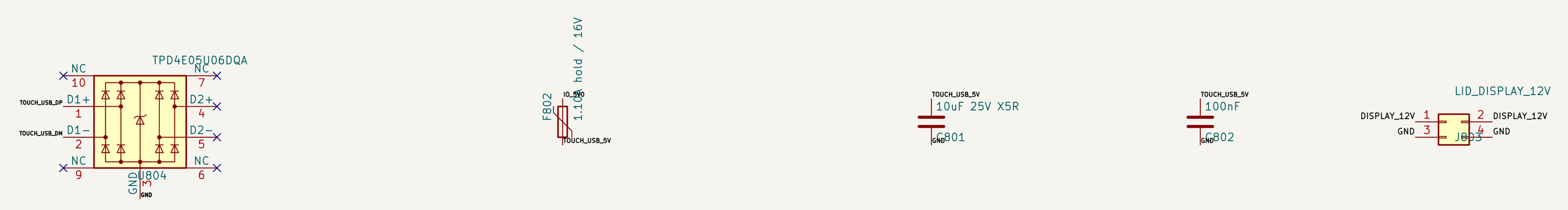
4. USB TOUCH HOST PORT



Only USB 2 is allocated from the CM5. A standard USB 3 A-to-B cable still carries USB 2 touch data.

The monitor-side SuperSpeed Type-B receptacle is not duplicated as a device connector on this host carrier.

5. 12 V / 2.5 A DISPLAY POWER



DISPLAY_12V design load: 2.5 A continuous / 30 W branch, covering the locked 25 W monitor.

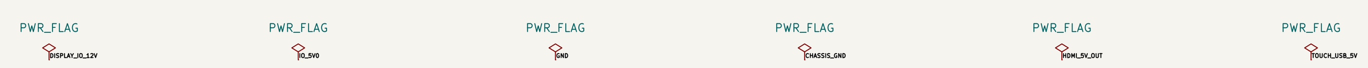
No local display eFuse. Use the upstream AUX_12V branch fuse and H03C keyed 18 AWG harness.

6. HINGE-EDGE HARNESS BUILD

H03A HDMI, H03B USB touch, H03C 12 V: 1000 +/- 25 mm each, with separate retained hinge and service loops.

Acceptance: full lid travel plus 300 mm panel lift / 45 degree tilt; no connector tension or hinge-line bend.

Carrier connectors face down. Store released loops clear of fans, PSU airflow, board keepouts and sharp edges.



No display eFuse; separate fused DISPLAY_12V and DISPLAY_IO_12V branches feed this sheet
USB-A is the CM5 host end; cable terminates in USB-B SuperSpeed at the monitor
Carrier connectors face downward at the hinge-edge harness notch
15.6-inch lid monitor: HDMI video, USB 2 touch, 12 V / 2.5 A

ProComm

Sheet: /Display Harness/
File: Display-Harness.kicad_sch

Title: Radxa CM5 ProComm Carrier — Lid Display Harness

Size: A1	Date: 2026-08-14	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 7/17

1. I2S0/TDM PROGRAM-AUDIO LVDS LINK

J901 mates with AUDIO-8X8 J101. Route five 100 ohm differential pairs with the assigned interleaved returns.

SN65LVDT2D includes the ADC-data receiver termination; do not add a duplicate far-end resistor.

2. I2S1 HEADSET LEVEL TRANSLATION

CM5 audio GPIO is 3.3 V. ES8316 digital pins remain entirely in the translated 1.8 V domain.

3. DEDICATED LOW-NOISE HEADSET RAILS

R900 is the only headset analog-ground bond. Do not create a second PCB or chassis bond.

4. ES8316 I2S1 HEADSET CODEC

CE is pulled to 1.8 V; all ES8316 digital pins are translated. Decoupling follows the Radxa reference topology.

5. TPA6132A2 HEADPHONE AMPLIFIER

G0=1 and G1=0 select 0 dB gain. EN remains software-controlled for pop-free mute.

6. CTIA HEADSET JACK, MIC BIAS, DETECT, AND ESD

CTIA contact map: tip=L, ring1=R, ring2=GND, sleeve=MIC. Verify isolated switch polarity on a Kycon sample.

J910 is routable for prototype capture, but its drawing-derived footprint is blocked from production until sample/coupon sign-off.

7. HEADSET CONTROL POLICY

HEADSET_POWER_EN and HEADSET_AMP_EN default low so the codec and amplifier remain muted during boot.

Sequence: enable rails, configure ES8316, enable TPA6132A2, then ramp software volume.

Firmware must debounce HS_JACK_DET_N and perform pop-free mute before any route change.

Kycon STX-353K7A is CTIA-mapped; its preliminary land pattern requires a coupon
TPA6132A2 provides 0 dB dedicated stereo headphone amplification
I2S1 uses explicit 3.3 V to 1.8 V translation before the ES8316 codec
I2S0/TDM program audio is buffered over LVDS to the separate AUDIO-8X8 board
ProComm

Sheet: /Headset / Audio Control/
File: Audio-Control.kicad_sch

Title: Radxa CM5 ProComm Carrier — TDM and Headset Audio

Size: A0	Date: 2026-08-16	Rev: A1
KiCad E.D.A. 10.0.5-rc1		Id: 8/17

Component values are Rev A1 starting values; bench loop, thermal, and transfer validation remain ma
DISPLAY_12V is a fused 12 V / 2.5 A harness branch with no dedicated eFuse
The 27.2 mF / 50 V no-blink bank resides on floor-mounted PWR-SELECT
RAW_OUT_LOAD accepts 10.5 V to 30 V after source selection and load telemetry

ProComm

Sheet: /Power Regulators/
File: Power-Regulators-A1.kicad_sch

Title: Radxa CM5 ProComm Carrier — Power Regulators and Sequencing

Size: A0	Date: 2026-08-14	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 9/17

TCA9535 0x20 captures source, fan, temperature, and power-monitor alerts
TMP117 sensors monitor CM5, modem, and board/power zones at 0x48/0x49/0x4A
PWM pull-ups provide full-speed hardware fallback if the controller or firmware fails
EMC2305 controls CPU, modem, intake, and exhaust fans independently

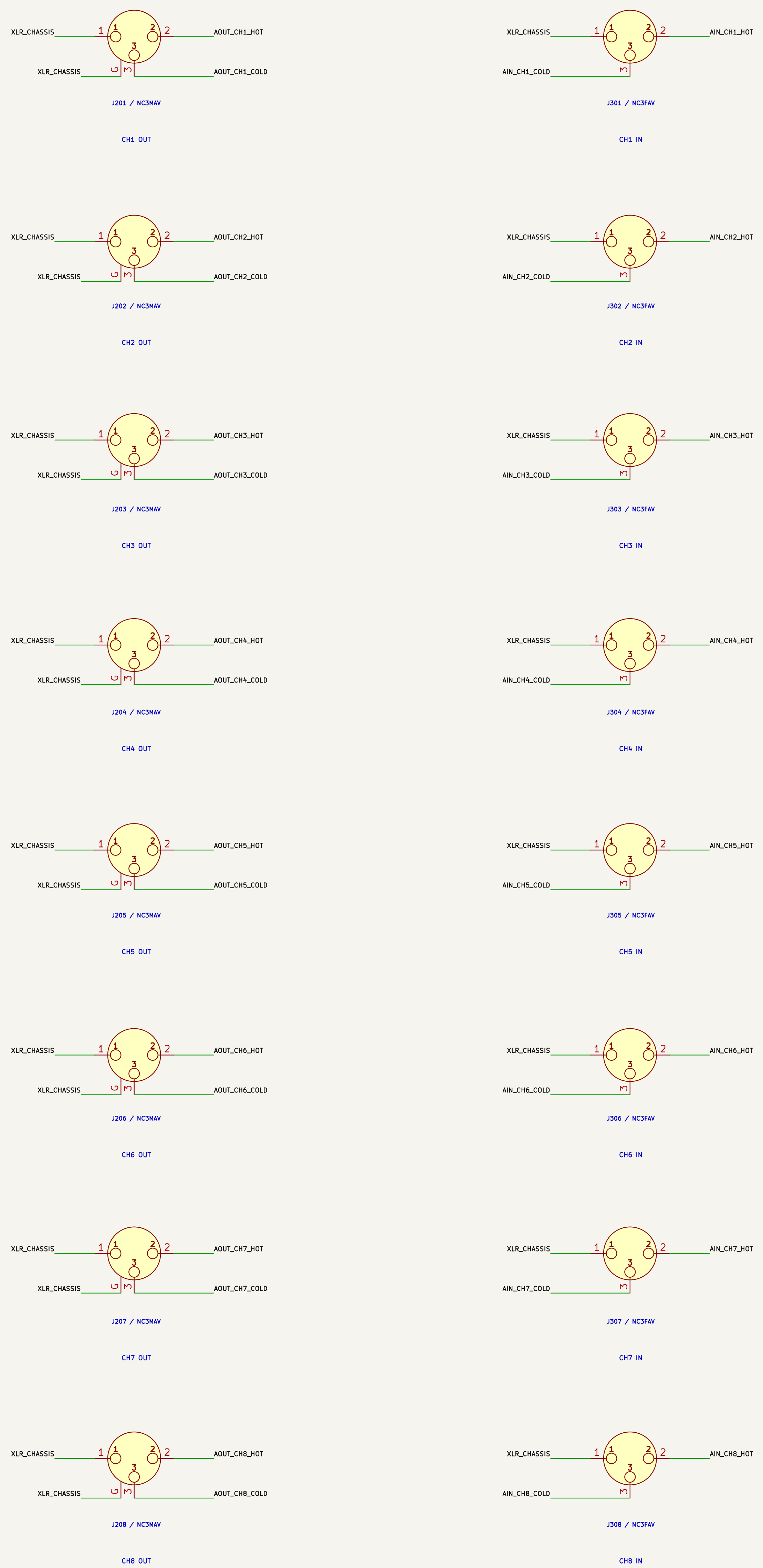
ProComm

Sheet: /Thermal / Fans / IO/
File: Thermal-IO.kicad_sch

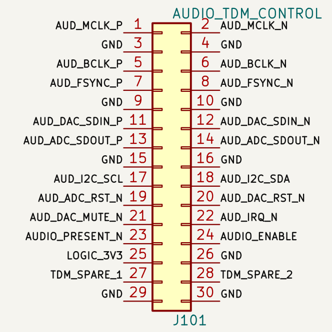
Title: Radxa CM5 ProComm Carrier — Thermal and GPIO Control

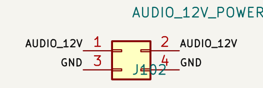
Size: A0	Date: 2026-08-14	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 10/17

A. PANEL–SUPPORTED BALANCED XLR BANK



B. CARRIER INTERFACES





Local filtered conversion produces +/−15 V, AKM 5 V analog and audio 3.3 V.

Mates with CM5–CARRIER J201; 100 ohm differential pairs with interleaved grounds.

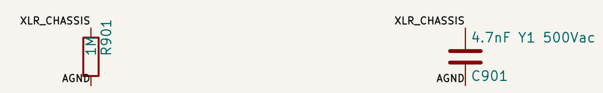
C. INPUT SIGNAL CHAIN

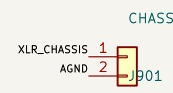
NC3FAV –> chassis/RFI/ESD –> THAT1206 –> level/filter network –> AK5558VN

Eight differential channels; no phantom power; +4 dBu nominal / +24 dBu max.

Detailed A1 sheet captures TDM256 straps, four VREFH feeds, low–noise rails, and the +24 dBu level map.

F. XLR SHIELD / CHASSIS BOND





Every XLR pin 1 and NC3MAV shell G bonds directly to the local chassis plane.

R901 and Y1–rated C901 form the single controlled RF/static bond to AGND; verify in EMC test.

No audio return current may use the XLR_CHASSIS copper or panel fasteners.

D. OUTPUT SIGNAL CHAIN

AK4458VN –> reconstruction/gain filter –> OPA165x –> THAT1646 –> RFI/fault network –> NC3MAV

Stable into 600 ohm test load; 10 kohm or higher is the normal operating load.

Hardware mute remains asserted through boot, clock loss, reset and power transfer.

E. DETAILED A1 CAPTURE SHEETS

01 Audio–TDM–Clock: LVDS TDM, protected I2C/control, clock distribution

02 AK5558–ADC: exact pins, TDM256/32–bit I2S straps, references and bypass

03 AK4458–DAC: exact pins, serial control, references and reconstruction contract

04 Audio–Inputs: 8x THAT1206, OPA1652, RFI/ESD and anti–alias networks

05 Audio–Outputs: 8x OPA1652/THAT1646, fail–silent relays and protection

06 Audio–Power: TRI20 +/−15 V, 5.5 V pre–rail, dual LT3045, 3.3 V and star

Routing hold: AKM exposed–pad coupons and TQ2 insertion/land–pattern coupon

ENGINEERING CAPTURE – detailed sheets complete; coupon and bench gates remain
Left bank male outputs; right bank female inputs
+4 dBu nominal, +24 dBu maximum, active–balanced XLR stages
AK5558VN ADC and AK4458VN DAC over buffered differential TDM

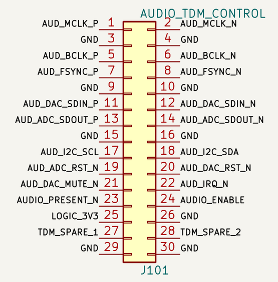
ProComm

Sheet: /AUDIO–8X8 Interface/
File: Audio–8x8.kicad_sch

Title: Radxa CM5 ProComm Audio 8x8 – Interface Contract

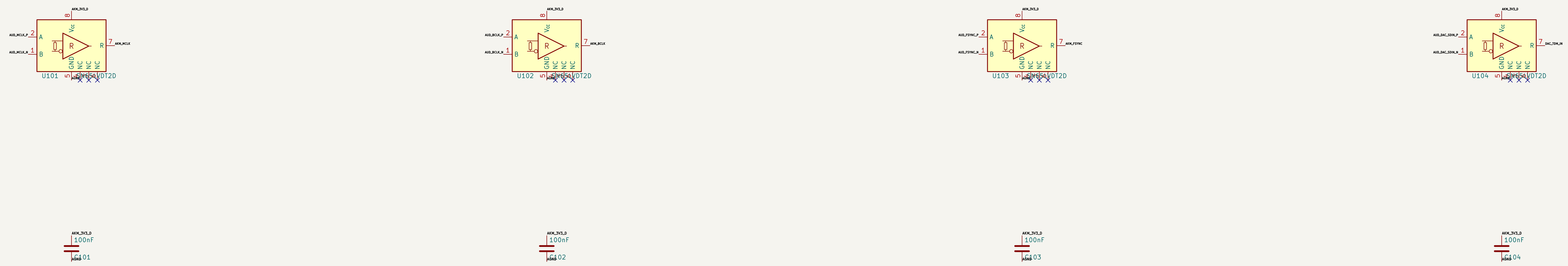
Size: A2	Date: 2026–08–16	**Rev: A1**
KiCad E.D.A. 10.0.5–rc1		Id: 11/17

1. LOCKED CARRIER HARNESS



Harness length and pair assignment are controlled by docs/audio_tdm_harness_a1.csv.

2. TERMINATED LVDS RECEIVERS



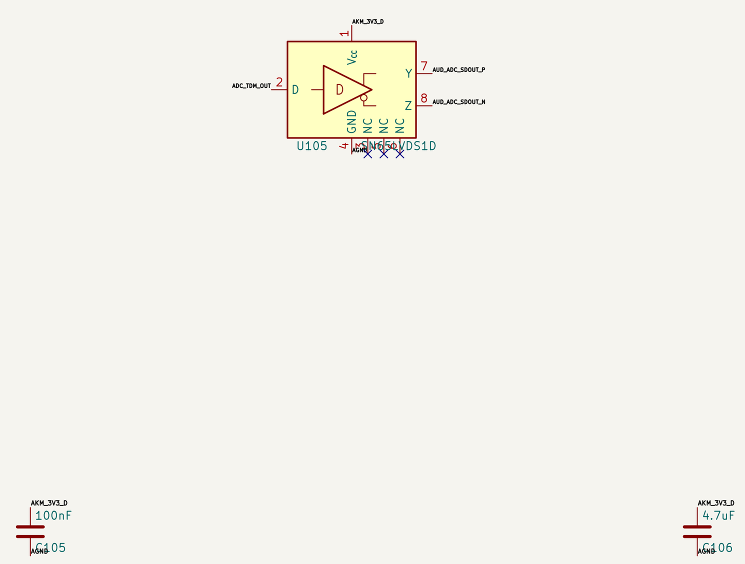
MCLK: Integrated 110 ohm termination

BCLK: Integrated 110 ohm termination

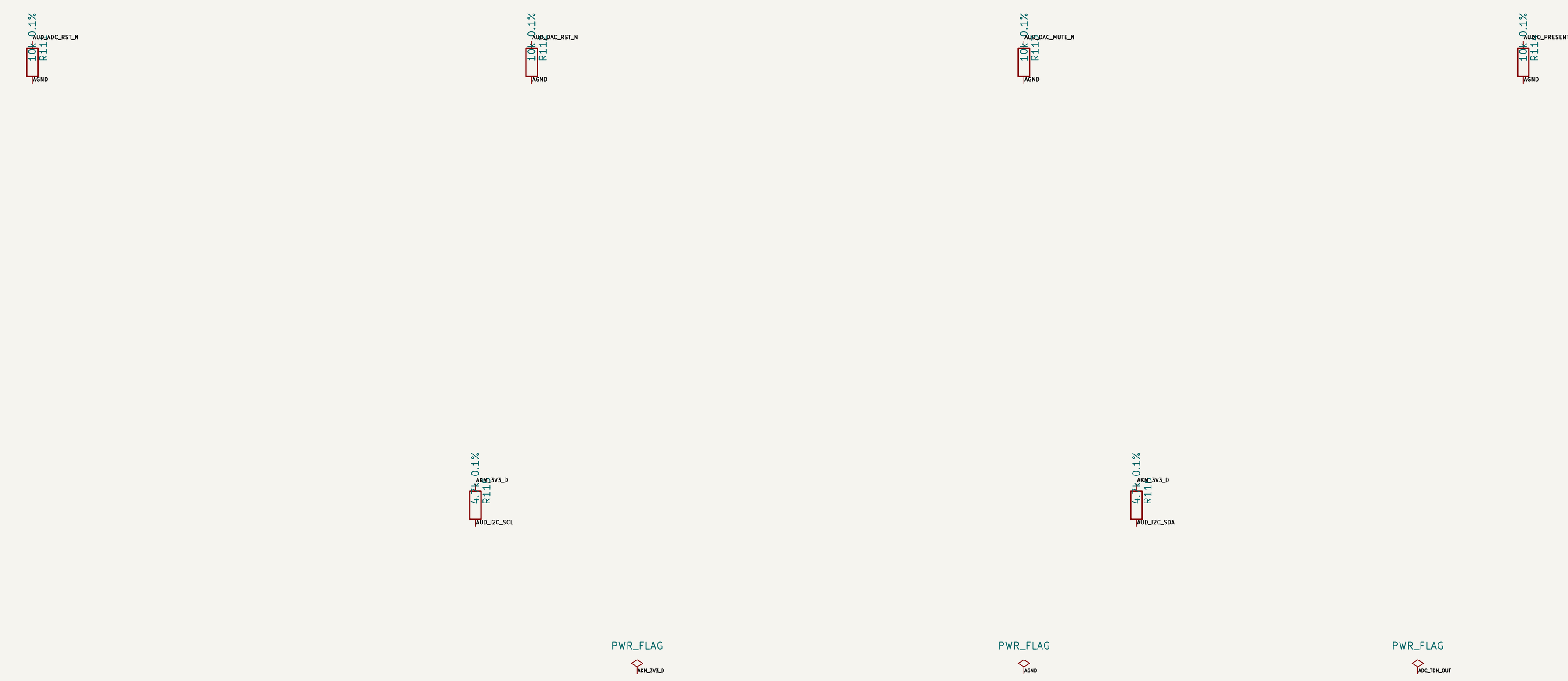
FSYNC: Integrated 110 ohm termination

DAC_SDIN: Integrated 110 ohm termination

3. ADC RETURN LVDS DRIVER



4. CONTROL DEFAULTS AND POWER STATE



Reset and mute default asserted while the harness is absent or the carrier is unpowered.

AUDIO_ENABLE is consumed by the local audio–power sequence; firmware releases resets only after all PG signals and clocks are valid.

Route AKM_MCLK, AKM_BCLK, AKM_FSYNC, DAC_TDM_IN, and ADC_TDM_OUT as short referenced single–ended traces after the translators.

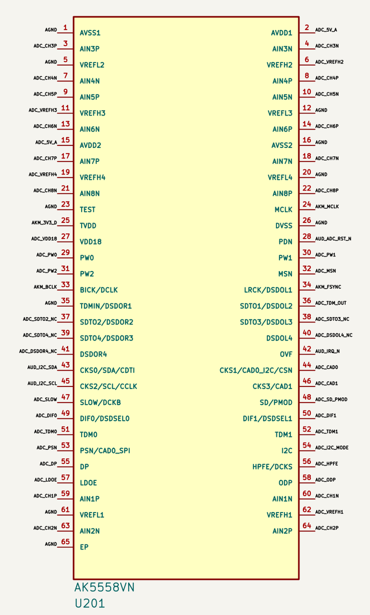
100 ohm differential; interleaved returns; no branch stubs
87832–6423 pin assignment is locked and matches CM5–CARRIER J901
One LVDS driver returns the ADC TDM data stream to the CM5 carrier
Four terminated LVDS receivers carry MCLK, BCLK, FSYNC, and DAC TDM data
ProComm

Sheet: /TDM / Clock / Control/
File: Audio–TDM–Clock.kicad_sch

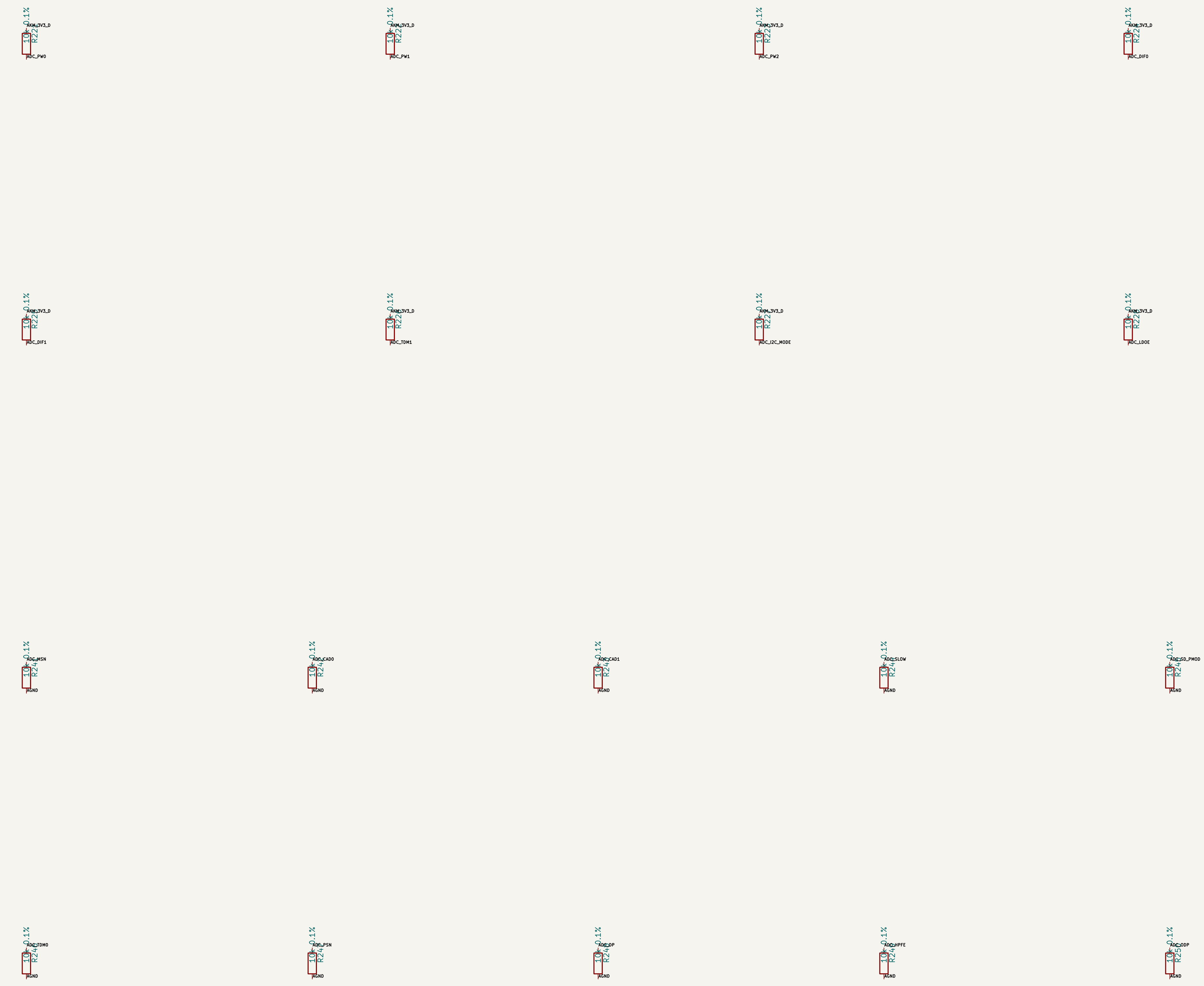
Title: ProComm AUDIO–8X8 – TDM, Clock, and Control Interface

Size: A1	Date: 2026–08–16	**Rev: A1**
KiCad E.D.A. 10.0.5–rc1		Id: 12/17

1. AK5558VN EXACT PIN CAPTURE

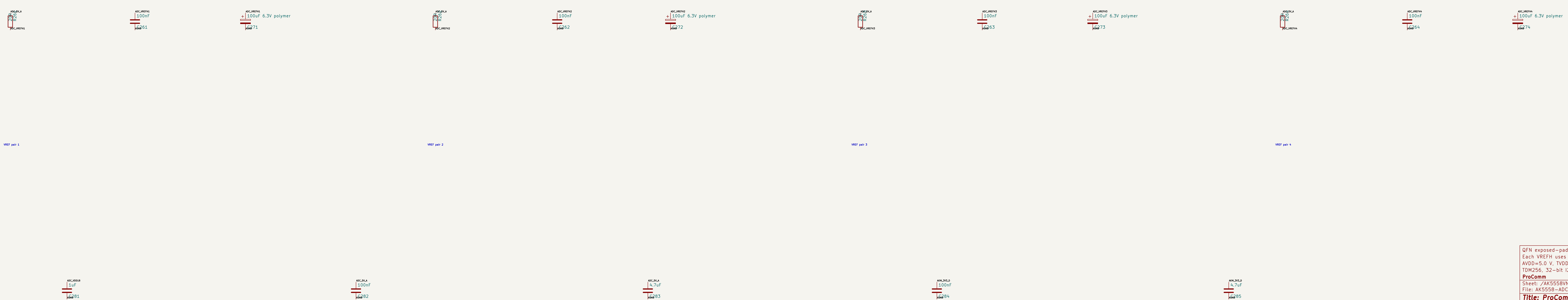


2. MODE STRAPS



Straps establish the safe hardware default; firmware still writes and verifies the serial–control registers after reset.

3. REFERENCES AND LOCAL BYPASS



QFN exposed–pad geometry and assembly coupon remain a routing release gate
Each VREFH uses 20 ohm plus 100 nF and 100 uF to its VREFL/AGND
AVDD=5.0 V, TVDD=3.3 V, internal 1.8 V LDO enabled
TDM256, 32–bit I2S, slave mode: TDM1:0=10, DIF1:0=11, MSN=0

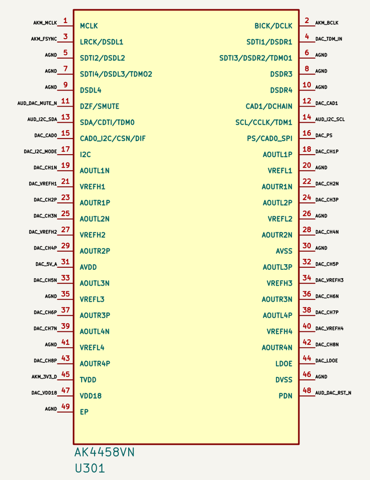
ProComm

Sheet: /AK5558VN ADC/
File: AK5558–ADC.kicad_sch

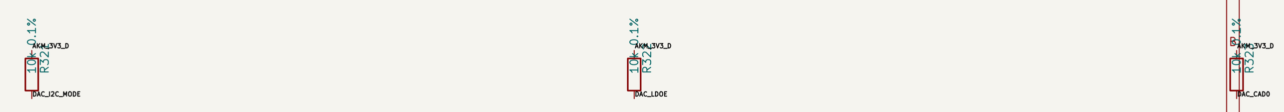
Title: ProComm AUDIO–8X8 – AK5558VN Eight–Channel ADC

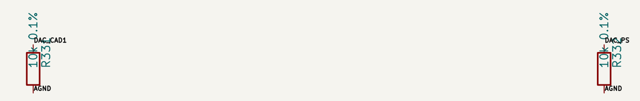
Size: A1	Date: 2026–08–16	**Rev: A1**
KiCad E.D.A. 10.0.5–rc1		Id: 13/17

1. AK4458VN EXACT PIN CAPTURE



2. SERIAL-CONTROL DEFAULTS

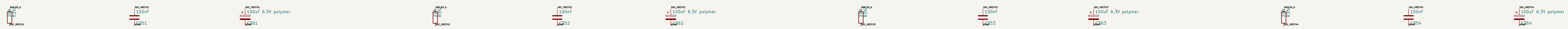




CAD1:0=01 selects DAC address 0x11; ADC remains at 0x10. TDM256 and 32-bit I2S are written only while mute is asserted.

The unused SDTI2/3/4 and DSD pins are tied low in PCM TDM mode to prevent floating inputs.

3. REFERENCES AND LOCAL BYPASS

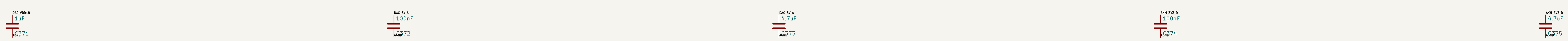


VREF pair 1

VREF pair 2

VREF pair 3

VREF pair 4



QFN exposed-pad geometry and assembly coupon remain a routing release gate
DAC mute remains asserted until rails, clocks, reset, and register readback pass
AVDD=5.0 V, TVDD=3.3 V, internal 1.8 V LDO enabled
TDM256, 32-bit I2S, slave mode is programmed and read back over I2C

ProComm

Sheet: /AK4458VN DAC/
File: AK4458-DAC.kicad_sch

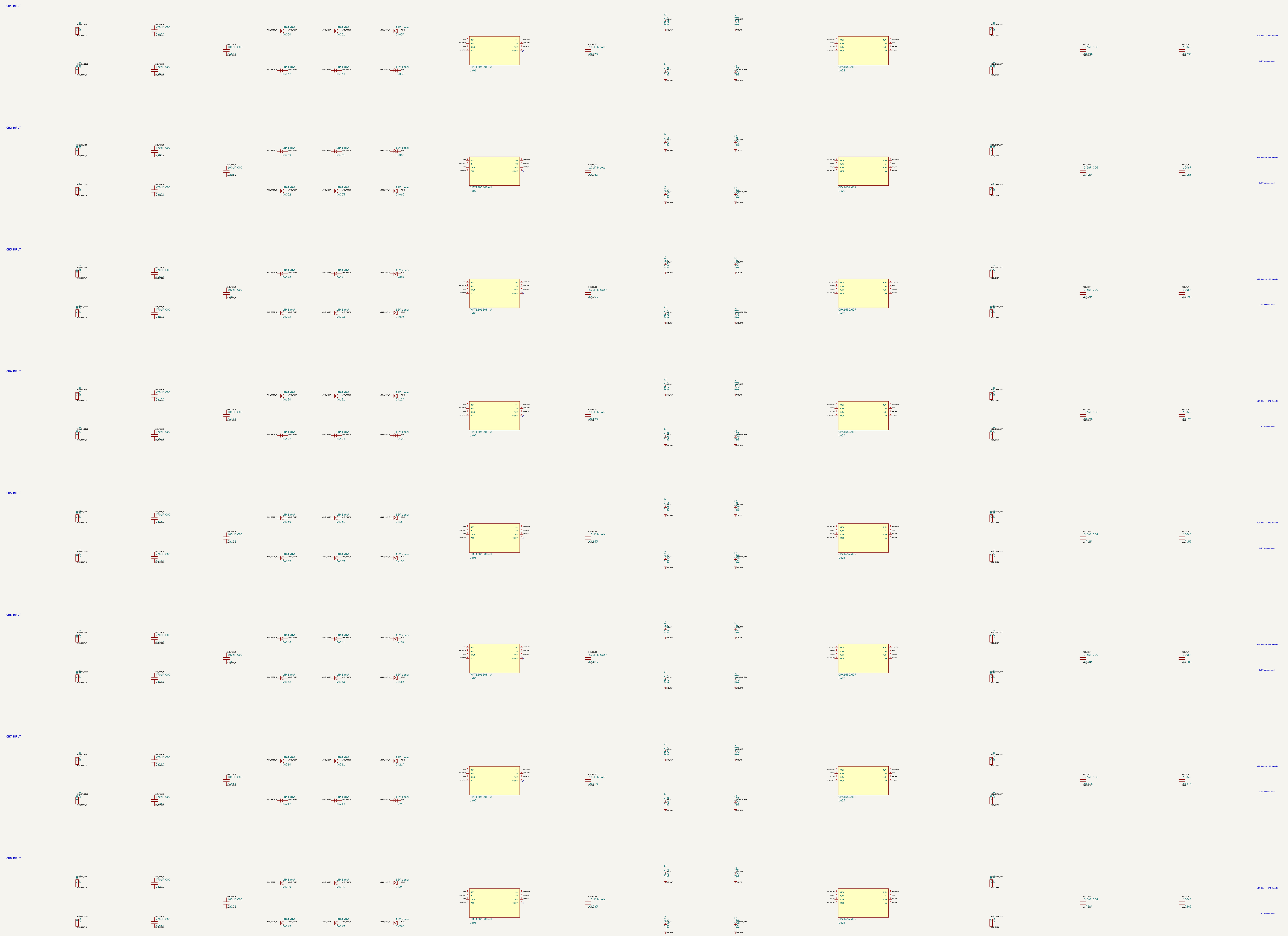
Title: ProComm AUDIO-8X8 – AK4458VN Eight-Channel DAC

Size: A1	Date: 2026-08-16	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 14/17

EIGHT IDENTICAL INPUT CHANNELS — SIGNAL FLOWS LEFT TO RIGHT

XLR -> 100R/RFI/fault clamps -> THAT1206 -> 10u bipolar -> 13k/1k level shift -> OPA1652 -> 220R/3.3nF -> AK5558

Every XLR pin 1 and shell bonds to XLR_CHASSIS at the connector. AGND and chassis meet only at the controlled board—entry bond.



INPUT RELEASE GATES: SPICE stability/noise, THAT fault—clamp review, +24 dBu THD+N, RF injection, ESD/surge, and phantom—fault tests on every connector type.

Protection and filter values require simulation plus surge/RFI bench qualification
13 k/1 k network maps +24 dBu to 2.49 Vpp differential at AK5558
+4 dBu nominal, +24 dBu maximum, no phantom power
THAT1206 −6 dB receiver; OPA1652 single—supply differential ADC driver

ProComm

Sheet: /Balanced Inputs 1—8/
File: Audio—Inputs.kicad_sch

Title: ProComm AUDIO—8X8 — Eight Balanced Input Channels

Size: A0	Date: 2026—08—16	**Rev: A1**
KiCad E.D.A. 10.0.5—rc1		Id: 15/17

EIGHT IDENTICAL OUTPUT CHANNELS — SIGNAL FLOWS LEFT TO RIGHT

AK4458 -> PCM reconstruction/difference amp -> x3 gain -> THAT1646 -> fail-silent relay -> ferrite/RFI/fault clamps -> XLR

Relays may energize only after all audio PG signals, TDM clocks, resets, register readback, and DAC unmute sequencing pass.

OUTPUT RELEASE GATES: AK4458/OPA SPICE, relay sample/insertion coupon, +24 dBu THD+N into 600 ohm and 10 kohm, short/open recovery, RF injection, ESD/surge, and power-transfer pop/click tests.

+24 dBu into 600 ohm is a qualification target; simulation and bench release required
TQ2-12V DPDT relays short both XLR legs to AGND while de-energized
OPA1652 gain stage and THAT1646 active-balanced line driver
AK4458 datasheet PCM LPF: 3.9 k, 4.7 k, 150 R, 470 pF, 3.9 nF

ProComm

Sheet: /Balanced Outputs 1-8/
File: Audio-Outputs.kicad_sch

Title: ProComm AUDIO-8X8 — Eight Balanced Output Channels

Size: A0	Date: 2026-08-16	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 16/17

1. AUDIO_12V ENTRY AND SINGLE GROUND STAR

R601 is a wide 2512 zero-ohm star: no fan, RF, Ethernet, or chassis current may share the AGND side.

AUDIO_12V source is separately fused on CM5-CARRIER; F601 protects this removable board and harness locally.

2. ISOLATED BIPOLAR LINE-STAGE RAILS

TRI20: 9-18 V input, +/-15 V at 670 mA per rail, 89% typical; place 4.7 uF directly at each output pin.

3. QUIET 5.5 V PRE-RAIL

TPS62913: 2.2 MHz low-noise mode; 29.4k/4.99k feedback; 470 nF NR/SS; verify ripple below the AKM rail budget.

4. SEPARATE ADC AND DAC 5 V LT3045 RAILS

ADC_5V_A: 4.99 V nominal, 300 mA limit, PG threshold about 4.5 V

DAC_5V_A: 4.99 V nominal, 300 mA limit, PG threshold about 4.5 V

5. AKM DIGITAL RAIL AND 2.5 V INPUT COMMON MODE

VCM_2V5 is a quiet divider reference for the eight ADC input op-amps only; no external load is permitted.

6. ERC SOURCES AND SEQUENCING CONTRACT

Release order: AUDIO_ENABLE -> 5V5_PG -> LT3045 PGs + AKM 3V3 valid -> clocks -> resets -> DAC unmute -> output relays.

Any PG loss immediately de-energizes all output relays and reasserts DAC mute before software intervention.

Power-good, sequencing, thermal rise, ripple, and conducted-emissions tests gate release
Separate LT3045 5 V/300 mA rails feed ADC and DAC; TPS7A2033 feeds AKM digital I/O
TRI 20-1223 creates +/-15 V; TPS62913 creates the quiet 5.5 V pre-rail
AUDIO_12V is the only carrier power input; local 3 A fuse and one GND-to-AGND star

ProComm

Sheet: /Audio Power/
File: Audio-Power.kicad_sch

Title: ProComm AUDIO-8X8 — Local Power, Sequencing, and Ground Star

Size: A1	Date: 2026-08-16	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 17/17